Experiment Design: Free-Shipping Threshold $100 → $75

*Status: drafted before analysing the experiment (see the process note in section 1). Planning inputs come from a separate historical sample of 150,000 control-experience users. All data in this project is **simulated**.*

1. Product brief

Feature	Free-shipping threshold at checkout: lower from $100 to $75, for every order on web and app
Who decides	Checkout / Growth PM proposes, VP Growth decides, Finance signs off on the margin definition
User problem (assumed)	Carts under the threshold see a $7.50 shipping fee late in checkout. Shoppers with a $75–$100 cart either leave or pad the cart to reach free shipping.
Hypothesis	Removing the fee for $75–$100 carts raises conversion enough to pay for the shipping cost we absorb on those orders
Why it is not obvious	Every order in the $75–$100 band becomes less profitable even if more people buy, so conversion alone would score the feature as a win
Decision	Roll out at $75 to everyone, roll out to a segment, or keep $100
Reversibility	A threshold is a config change, so a wrong launch can be rolled back. The cost of being wrong is the margin given away until someone notices
What the decision is worth (planning figure, all traffic)	About 10.4M visitors a year × $1.06 margin per visitor ≈ $11.1M margin a year. 1% of margin per visitor ≈ $111K a year; the 6.7% MDE ≈ $739K a year

Metric tree

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Margin per visitor  =  conversion rate  ×  margin per order
margin per order    =  42% × basket × (1 − return)  −  net shipping cost  −  $6 × return
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Decision rule (evaluated on the primary population after validity checks):

Outcome	Condition	Action
Ship to everyone	Conversion significantly up (Holm $p < 0.05$) **and** margin per visitor 95% CI lower bound above −2% **and** return rate not significantly up	Roll out; keep a 5% holdout
Do not ship broadly; targeted follow-up	Anything else that is not a kill, i.e. margin is inconclusive	Test the pre-registered segment (new customers) or a cheaper variant
Keep $100	Margin CI upper bound below 0, **or** conversion not significantly up, **or** return rate significantly up	Stop

The −2% is the margin loss Growth would tolerate in exchange for new-customer growth. In a real project I would agree it with Finance before the test.

Process note: in this portfolio build, the first draft already contained the margin criterion (CI lower bound above −2%) and the rule "inconclusive margin means do not ship broadly". The Keep-$100 row, the return-rate wording and the table layout were added afterwards for clarity. The recommendation does not depend on them.

2. Design

Item	Choice	Why
Randomisation unit	User (assigned on first visit in the window)	Avoids one person seeing both prices
Allocation	50 / 50	Maximises power
Window	14 days	Covers two full weekly cycles; business cap on run time
Traffic	about 28,571 users/day, 400,000 total, 200,000 per arm	Planning assumption
Exposure logging	Client-side tag	**Known risk:** tag failures would cause SRM, so SRM is checked first

3. Metrics

Role	Metric	Definition
Primary	Margin per visitor	42% × merchandise value after returns − net shipping cost − $6 per return, summed per assigned user (zeros included)
Driver (secondary)	Conversion rate; net revenue per visitor	Did the feature do what it was built to do?
Mechanism (diagnostic)	AOV; items per order; net shipping cost per order	Explain *why* margin moved. Shipping cost rises by construction, so it cannot be a guardrail
Guardrail	Return rate per order	Extra buyers may return more. Page-load time is not

		simulated
Pre-registered segments	New vs returning; device	Chosen *before* data, because new customers are expected to respond most
Exploratory segments	Traffic source	Reported but flagged exploratory

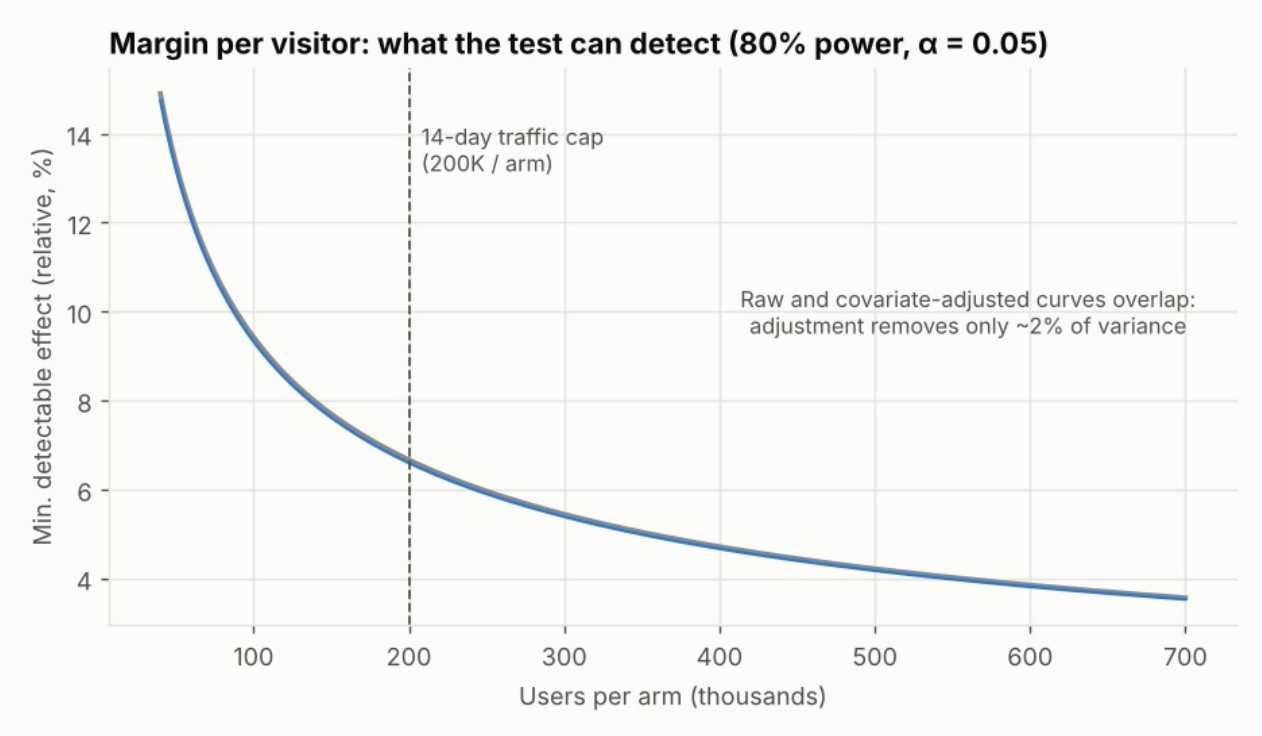
Assumptions (cost model): carrier cost $9.00 per order; customers under the threshold pay a $7.50 fee, so net shipping cost is $1.50 (paid) or $9.00 (free).

4. Baseline and variance (from the historical sample)

- Conversion: **3.24%**
- Margin per visitor: **$1.06**, standard deviation **$8.00** (coefficient of variation 7.5: zero-inflated and heavy-tailed)
- Outlier rule: winsorise per-user margin at the pilot 99.95th percentile (**$121.94**); report raw and winsorised results

5. Power and sample size (α = 0.05 two-sided, power 80%)

Question	Answer
Users per arm needed for a **5%** relative lift in margin per visitor	**355,011** → about **25 days** at planned traffic
Users per arm available in 14 days	**200,000**
Min. detectable effect on margin per visitor in 14 days	**6.7%** relative (6.6% after covariate adjustment)
Min. detectable effect on conversion in 14 days	**4.9%** relative
Simulation check of power at the margin MDE	formula says 80%, bootstrap simulation says **76%** at a 6.7% effect (heavy tails make the formula optimistic)



MDE vs sample size

Honest planning conclusion: a 14-day test is well powered for conversion but **under-powered for margin per visitor**. We accept this because 14 days is the business cap, and we record it up front: if margin per visitor comes back inconclusive, that is the expected outcome, not a failure of the test. Covariate adjustment helps little (2.1% variance reduction in the pilot) because a purchase is a rare event and 40% of users are new with no history.

6. Analysis plan (fixed in advance)

1. **Validity first:** SRM chi-square at α = 0.001; covariate balance (|SMD| > 0.1 flagged); 1,000-split A/A test of the pipeline.
2. **Primary:** difference in mean margin per visitor with 95% CI (Welch), cross-checked by bootstrap, winsorised, CUPED and regression-adjusted versions.
3. **Driver, mechanism and guardrail metrics:** Holm-adjusted p-values across the six non-primary metrics; ratio metrics by delta method (users are the randomisation unit).
4. **Segments:** pre-registered segments get a formal interaction test; all segment cells are BH-adjusted.
5. **Time effects:** week-1 vs week-2 comparison and a linear trend test on the treatment effect.
6. **Decision rule:** the decision table in section 1.
7. **Looks:** no early stopping. A sequential design is evaluated separately (see peeking analysis).